

THE TRANSFORMATIVE POWER OF GEOAI: A CASE OF STUDY FOR SMALL FARMERS' EUDR COMPLIANCE

DIPLOMA THESIS
OLOMOUC 2026



RESEARCH PROBLEM

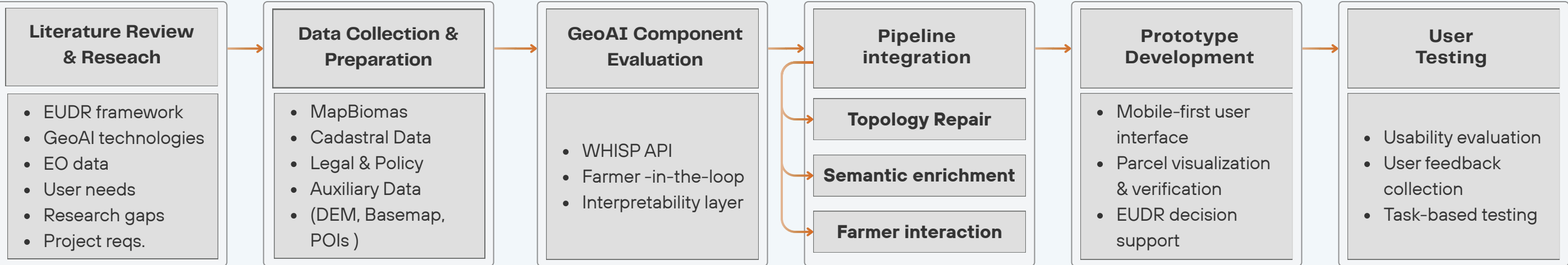
How can GeoAI-powered digital tools bridge the gap between EUDR compliance requirements and smallholder farmers' demands?

The EUDR requires plot-level geolocation and deforestation evidence. Smallholders often face inaccurate boundaries, fragmented data and limited technical capacity. This research connects automated spatial validation, Earth Observation and farmer knowledge through an explainable, user-in-the-loop GeoAI workflow.

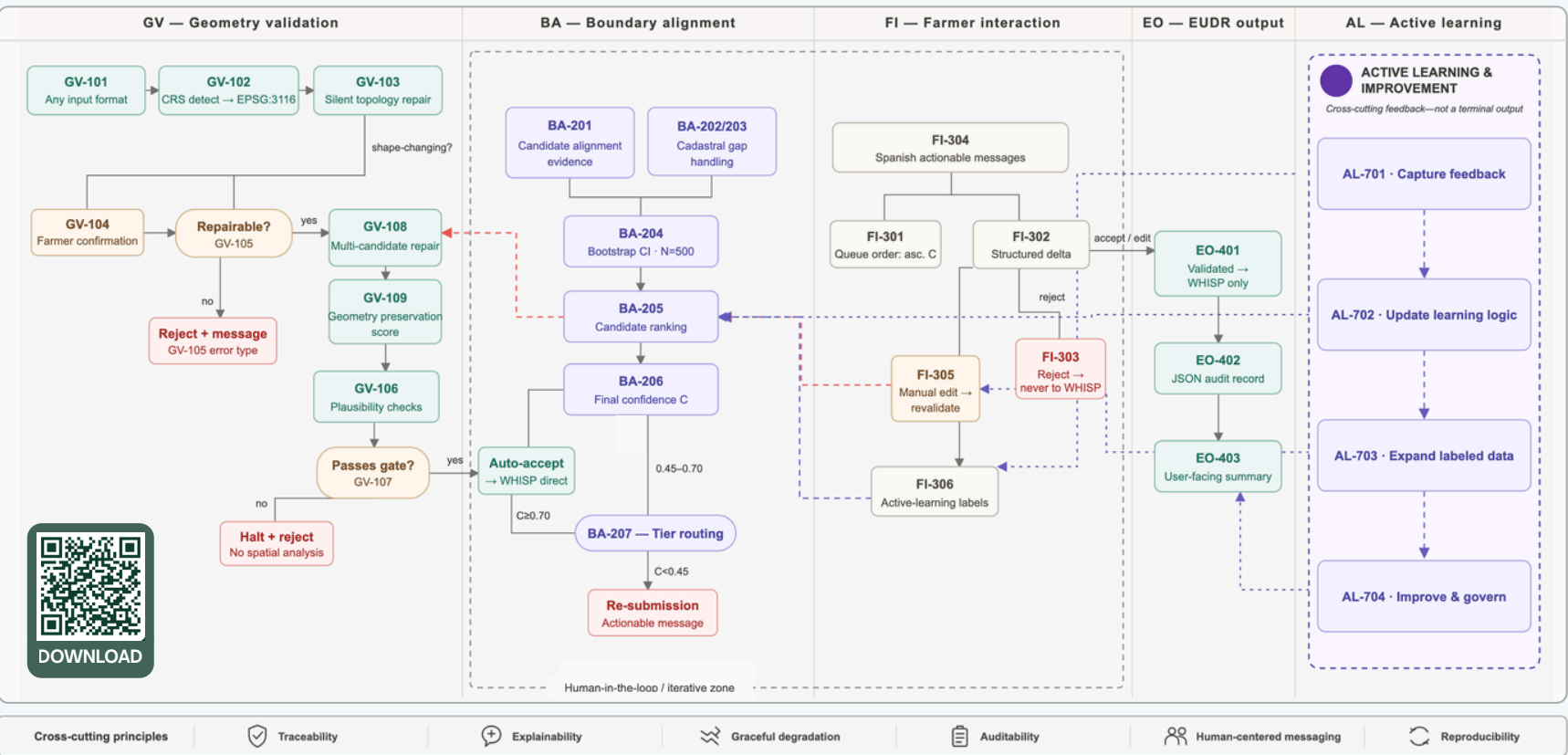
OBJECTIVES

- 01 Design a replicable and scalable **farmer-in-the-loop GeoAI pipeline** for parcel harmonization and topology repair to support EUDR compliance for small-farmers.
- 02 Translate pipeline into an **interactive dashboard** integrating GeoAI to support parcel validation and deliver interpretable EUDR compliance insights.
- 03 Evaluate and document the system through technical testing, active-learning records, and structured user studies, assessing its usability and transparency.

METHODOLOGY

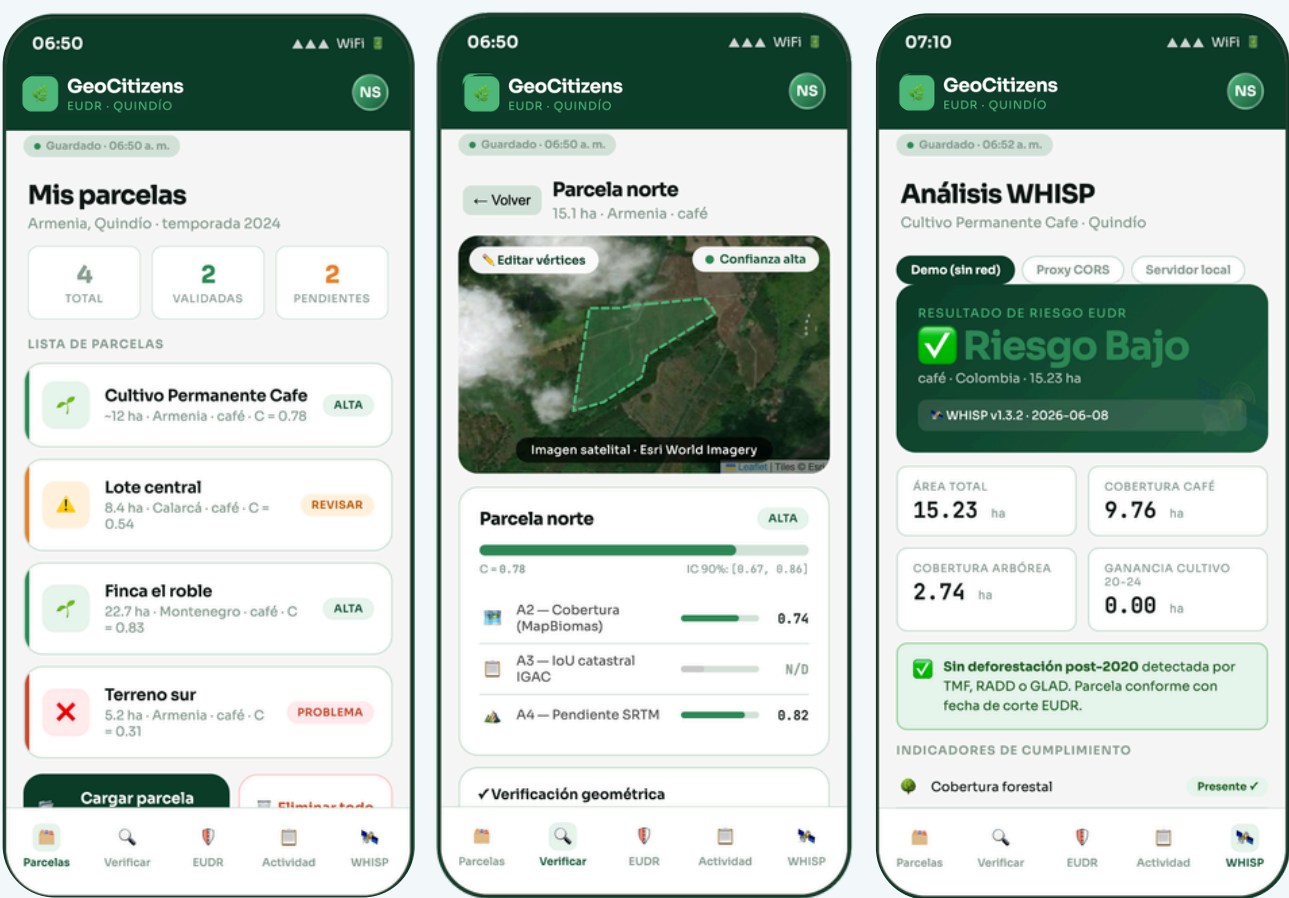


RESULTS - PIPELINE



- P0 Harmonization**
Standardize schemas, CRS, units & scales.
- P1 Topology Repair**
Fix geometry errors & ensure topology.
- P2 Active Learning Farmer Queue**
Prioritize uncertain parcels for review.
- P3 WHISP Model Integration**
Integrate deforestation probability & alerts.
- P4 Additional Evidence**
Add complementary datasets & contextual signals.

RESULTS - DASHBOARD



- Detect invalid geometries**
Self-intersections, spikes & duplicate vertices before compliance assessment.
- Generate repair & boundary-alignment candidates.**
- Rank candidates with available geospatial evidence**
Missing layers are excluded and weights redistributed.
- Route ambiguous or low-confidence cases to the farmer**
- Preserve confirmations, edits & rejections in an audit log.**

CONCLUSION

Human-centred GeoAI can make regulation legible to smallholder farmers by translating complex regulatory requirements and geospatial evidence into accessible and transparent information. This approach strengthens farmers' participation and decision-making while providing an inclusive and scalable pathway for EUDR compliance.

IMPACT



Contribution to UN sustainable development goals (SDG)



AUTOMATE WHAT IT'S CERTAIN, EXPLAINS WHAT IS NOT

